Computer Networks

Student Hub | Original study summary | For revision and self-study

Course

CS304 | Student Hub Original Revision Guide

1. Layering

Layering separates responsibilities. The OSI model has physical, data-link, network, transport, session, presentation, and application layers. TCP/IP groups these into link, internet, transport, and application layers.

2. Ethernet and IP

Ethernet delivers frames within a local link using hardware addresses. IP provides logical addressing and best-effort packet forwarding across networks. Routers use destination prefixes to choose a next hop.

3. Subnets and routing

A subnet mask or prefix length identifies the network portion of an address. Routing tables choose matching prefixes, generally preferring the most specific match. ARP or neighbor discovery resolves a local next-hop address.

4. Transport protocols

TCP provides a reliable ordered byte stream using sequence numbers, acknowledgements, retransmission, and congestion control. UDP provides datagrams without built-in delivery guarantees.

5. Common application protocols

DNS maps names to records. HTTP transfers web requests and responses. TLS protects application traffic with encryption and server authentication when configured correctly.

6. Troubleshooting

Work from the nearest layer outward: link, IP configuration, route, DNS, transport port, then application. Record the exact error and compare a working host or service before changing settings.

Study checklist

Draw how a browser request travels through layers, calculate a simple network prefix, compare TCP and UDP, and explain how DNS fits before a typical web request.

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Practice prompts

1) Which layer routes IP packets? 2) What does TCP add that IP does not promise? 3) What is DNS used for? 4) Why does a router use a subnet prefix?

Suggested answers

1) The network or internet layer. 2) Reliable, ordered delivery between endpoints. 3) Resolving domain names to records such as IP addresses. 4) To determine which destination network and next hop match a packet.

Troubleshooting habit

Use one test at a time and keep the result: link status, IP address, gateway reachability, DNS lookup, and service connection. Avoid changing several variables at once.

Further reading

Cisco Networking Academy and Cloudflare Learning Center links are listed on the Student Hub Course Materials page.